Open Source for Regulated Manufacturing

GMP-Ready IIoT Data Logging with Apache IoTDB

Lukas Ott otluk@apache.org

The Pressure: Connected Manufacturing Meets Regulation

The Shopfloor Is Becoming a Data Platform

The Core Tension

- Manufacturing needs more real-time data.
- GMP requires trustworthy and inspectable records.
- AI and analytics increase the demand for structured, contextualized data.
- Legacy shopfloor architectures are often fragmented and opaque.

The future shopfloor needs data platforms that are both scalable and inspectable.

Regulatory Pressure Is Expanding

Major Aspects

The Regulatory Lens

Part 11 vs. Annex 11 in 60 Seconds

The Shared Compliance Goal

Data Integrity Becomes Architecture

ALCOA++ as the Practical Lens

ALCOA++ Translated to IIoT

For IIoT, ALCOA++ is not a documentation checklist. It is an architecture requirement.

Attributable: Every reading is linked to a machine, sensor, gateway, service, and timestamp.	**Contemporaneous**: Data is captured **immediately** at the machine event.
Original: Raw values are preserved **before** any transformation or aggregation.	**Complete**: Retries, gaps, corrections, and synchronization events remain **traceable**.

Compliancy and Data

What Can Go Wrong on the Shopfloor?

Network outage during production	Duplicate readings after retry
Missing or inconsistent timestamps	Unauthorized data access
Silent data loss during synchronization	Uncontrolled transformation logic
Retention or backup failure	Configuration drift between environments

A GMP-ready IIoT architecture must be designed for failure, not only

Why Open Source?

Why Open Source Helps Validation

Open Source Is Not Automatically Compliant

- The software is not validated by default.
- The implemented system must be qualified and validated.
- Supplier and community governance still need assessment.
- Security patching and dependency management must be controlled.
- Documentation, testing, SOPs, and change control remain mandatory.

Open source is not a shortcut around GMP. It is a transparent foundation that can support validated architectures.

What to Remember About Open Source

From Open Architecture to Apache IoTDB

Open Architecture for Audit-Ready History

Example Intended Use

- Capture high-frequency machine and sensor readings.
- Buffer data locally when central connectivity is unavailable.
- Synchronize data into a historian without duplicates.
- Visualize current and historical data.
- Support traceability, operational resilience, and inspection readiness.

The goal is not to validate a database in isolation. The goal is to validate the implemented system for its intended use.

Apache-Based IIoT Data Logging Reference Architecture

Apache IoTDB as the Historian Core

- Optimized for high-frequency industrial time-series data.
- Stores structured device and sensor measurements.
- Supports integration with authentication, authorization, encryption, monitoring, and retention controls.
- Can serve as the historian core inside a validated IIoT architecture.
- Works together with controlled ingestion, metadata, logging, and validation processes.

Apache TSFile as the Storage Foundation

- Time-series file format optimized for efficient storage and query.
- Supports append-oriented storage patterns.
- Helps separate raw time-series history from downstream analytics.
- Supports reproducibility and inspection when combined with controlled storage, access, and retention design.
- Can be paired with analytical formats such as Apache Iceberg.

Mapping Compliance Expectations to Technical Controls

From Architecture to Validation Evidence

Compliance and Validation

Validation Starts with System Boundaries

- What is GMP-relevant?
- What is the regulated record?
- Which components create, modify, store, or display the record?
- Which integrations are in scope?
- Which users, services, and systems can access the data?
- What is the intended use?
- What is explicitly out of scope?

Risk-Based Validation with GAMP 5

1. **Intended Use**: Define what the system is expected to do.
2. **Risk Assessment**: Identify patient, product quality, and data integrity risks.
3. **User Requirements**: Define business, quality, and operational requirements.
4. **Design Specifications**: Translate requirements into controlled architecture and configuration.
5. **IQ / OQ / PQ**: Prove installation, operation, and performance.
6. **SOPs and Training**: Ensure controlled and repeatable use.
7. **Monitoring and Review**: Verify ongoing compliant operation.

What Good Testing Looks Like

Test	What it proves	IIoT example
IQ	The system is installed repeatably and correctly.	Correct versions, configuration, certificates, storage paths, deployment scripts.
OQ	The system functions as specified.	RBAC, retention, backup, synchronization, failover, audit logging, monitoring.
PQ	The system performs under realistic production conditions.	Peak sensor load, network outage, replay, idempotent sync, long-running ingestion.

Part 11 Control Themes

The Demo: Resilience and Traceability

Demo Scenario: Network Outage Without Data Loss

- A production line continues generating readings.
- The central historian is temporarily unavailable.
- The local system continues accepting data.
- Readings are buffered safely.
- Synchronization resumes when connectivity returns.
- Duplicate records are avoided through idempotent behavior.
- Historical data becomes visible in Apache IoTDB.

The demo proves continuity, traceability, and recoverability under realistic shopfloor conditions.

Live Demo: Hybrid IoT Data Logging and Visualization

- Goal: show the path from ingestion to buffering to synchronization.

Demo Steps

Step	Action
1	Log in as operator with operator.
2	Send sample readings to /ingest or use the ingestion helper.
3	Open the dashboard and observe the local buffer chart.
4	Trigger Sync to IoTDB and watch the live SSE progress stream.
5	Verify that synchronized historical data appears in the IoTDB chart.

Conclusion

From Edge Data to Audit-Ready History

Final Takeaways

- Regulated manufacturing does not just need more data.
- It needs data that can be trusted, inspected, and preserved.
- Apache IoTDB can serve as the historian core of a GMP-ready IIoT architecture.
- Apache technologies provide modular building blocks from edge to analytics.
- Validation depends on intended use, risk, evidence, procedures, and lifecycle control.
- Open source supports transparency, scalability, and validatability.

Open source is not a shortcut around validation. It is a foundation for transparent, scalable, and inspectable manufacturing data platforms.

Q&A

- Thank you.
- Projects:
  - Apache IoTDB: https://iotdb.apache.org/
  - Apache PLC 4 X: https://plc 4 x.apache.org/
  - Apache StreamPipes: https://streampipes.apache.org/
- Contact:
  - Lukas Ott <otluk@apache.org>